

COMPREHENSIVE CURATED WORKSHEET

Chapter 1: Integers — Exercises 1A to 1C

Prepared for Students of Class VII Mathematics — Maximum Marks: 50

Worksheet Scope: This worksheet is carefully curated to assess and reinforce concepts from the beginning of the chapter up to Exercise 1C. Topics include: representation of integers on a number line, absolute value, comparison, successor/predecessor, fundamental operations (addition, subtraction, multiplication, division), and their associated mathematical laws (closure, commutativity, associativity, and distributivity).

SECTION A: MULTIPLE CHOICE QUESTIONS (10 Marks)

Answer all questions. Each question carries 1 mark. Select the most appropriate option.

1. If a and b are two integers such that a is the predecessor of b , then the value of $a - b$ is:
(a) 1 (b) -1 (c) 0 (d) 2
2. Which of the following statements is true for any two negative integers x and y ?
(a) $x + y > 0$
(b) $x - y$ must be negative
(c) $x \times y > 0$
(d) $x \div y$ is always an integer
3. Which of the following is the multiplicative identity for the set of integers?
(a) 0 (b) -1 (c) 1 (d) None of these
4. If $(-12) \times (-5) \times a = -120$, then the value of the integer a is:
(a) 2 (b) -2 (c) 10 (d) -10
5. The predecessor of the successor of -15 is:
(a) -14 (b) -16 (c) -15 (d) -13
6. On a horizontal number line, if an integer p lies to the left of another integer q , then which of the following is correct?
(a) $p > q$ (b) $p < q$ (c) $p = q$ (d) $p = -q$
7. The value of $|-25| - |12| + |-5|$ is equal to:
(a) -8 (b) 18 (c) 8 (d) -18
8. A student verified that $(-8) \times \{5 + (-3)\} = (-8) \times 5 + (-8) \times (-3)$. This identity illustrates which algebraic law?
(a) Commutative Law of Multiplication
(b) Distributive Law of Multiplication over Addition
(c) Associative Law of Multiplication
(d) Closure Property of Addition

HOTS MCQ 1 (High Order Thinking Skills)

Let a be a negative integer and b be a positive integer. Which of the following expressions **must** always represent a negative integer?

9. (a) $b - a$ (b) $a - b$ (c) $a^2 + b$ (d) $|a| - b$

HOTS MCQ 2 (High Order Thinking Skills)

If $a \times b = -12$ and the sum $a + b$ is a positive integer, which of the following could be the value of $a - b$? (Assume a and b are integers).

10. (a) 7 (b) -7 (c) 13 (d) Both (a) and (c)

SECTION B: SUBJECTIVE PROBLEMS (30 Marks)

Solve all problems. Show complete workings and mention the mathematical rules or properties applied. Each problem carries 3 marks.

1. **Basic Arithmetic Computation:** Evaluate the following expression by applying the correct signs rule:

$$(-24) + 45 - (-16) - 13 + (-8)$$

2. **Absolute Value Evaluation:** Evaluate the numerical expression:

$$|-45| \div |-9| \times (-3) - |12|$$

3. **Ascending Order & Intervals:**

- (a) Arrange the following integers in ascending order: $-18, 12, -4, 0, -25, 9, -1$
(b) Write all integers that lie strictly *between* the smallest and the second-smallest integers in the above list.

4. **Successor & Predecessor Operations:** Find the product of the predecessor of -12 and the successor of -7 .

5. **Algebraic Properties Verification:** Let $a = -6$, $b = -4$, and $c = 2$. Verify the Distributive Law of Multiplication over Addition:

$$a \times (b + c) = (a \times b) + (a \times c)$$

6. **Temperature Word Problem:** At 6:00 AM, the temperature of a high-altitude mountain station was -5°C . By 2:00 PM, the temperature rose by 12°C due to solar heating, and by midnight, it fell by 14°C due to a cold wind. Calculate the final temperature recorded at midnight.
7. **Commercial Word Problem:** A vendor earns a profit of Rs. 5 on each pen sold and incurs a loss of Rs. 2 on each pencil sold. If she sells 80 pens and 120 pencils on a particular day, find her net profit or loss for that day.
8. **Division with Unknowns:** The product of two integers is -144 . If one of the integers is -9 , find the other integer.

HOTS Problem 1 (High Order Thinking Skills)

Let x and y be integers such that $|x| \leq 4$ and $|y| \leq 3$. Based on these boundaries, determine:

9. (a) The maximum possible value of the expression $x - y$.
- (b) The minimum possible value of the product $x \times y$.

HOTS Problem 2 (High Order Thinking Skills)

An explorer is standing on a research platform 12 meters above sea level. Directly below him, a research submarine is floating at a depth of 45 meters below sea level (represented as -45 meters).

10. (a) If the submarine begins to ascend to the surface at a constant rate of 5 meters per minute, write an integer equation for its position relative to sea level, $d(t)$, after t minutes.
- (b) Calculate how long it will take for the submarine to reach a depth of -15 meters.
- (c) What will be the absolute vertical distance between the explorer on the platform and the submarine when the submarine reaches this depth of -15 meters?

SECTION C: COMPLETE STEP-BY-STEP SOLUTIONS

This section contains rigorous, mathematically structured answers for self-assessment and verification.

Solutions for Section A: Multiple Choice Questions

1. Correct Option: (b) -1

Proof: If a is the predecessor of b , then by definition, $a = b - 1$.

Subtracting b from both sides yields: $a - b = -1$.

2. Correct Option: (c) $x \times y > 0$

Proof:

- (a) The sum of two negative integers is always negative (e.g., $-1 + (-1) = -2 < 0$), so $x + y > 0$ is false.
- (b) $x - y$ can be positive if y is more negative than x (e.g., $-2 - (-5) = -2 + 5 = 3 > 0$).
- (c) The product of two negative integers (same sign) is always a positive integer (e.g., $(-2) \times (-3) = 6 > 0$). This statement is always true.
- (d) Division of integers is not closed; $x \div y$ may be a fraction (e.g., $-2 \div -3 = \frac{2}{3} \notin \mathbb{Z}$).

3. Correct Option: (c) 1

Proof: An integer e is a multiplicative identity if $a \times e = e \times a = a$ for all $a \in \mathbb{Z}$. Since $a \times 1 = a$, 1 is the multiplicative identity. (Note: 0 is the additive identity).

4. Correct Option: (b) -2

Proof: Simplify the given equation:

$$\{(-12) \times (-5)\} \times a = -120 \implies 60 \times a = -120$$

Dividing both sides by 60 :

$$a = -120 \div 60 = -2$$

5. Correct Option: (c) -15

Proof:

- First, find the successor of -15 : Successor $= -15 + 1 = -14$.
- Next, find the predecessor of this result: Predecessor $= -14 - 1 = -15$.

In general, the predecessor of the successor of any integer x is x itself.

6. Correct Option: (b) $p < q$

Proof: On a horizontal number line, integers increase as we move from left to right. Therefore, any integer situated to the left of another integer is always smaller than it. Thus, $p < q$.

7. Correct Option: (b) 18

Proof: First, evaluate the absolute values (modulus): $|-25| = 25$, $|12| = 12$, and $|-5| = 5$. Now substitute these into the expression:

$$25 - 12 + 5 = 13 + 5 = 18$$

8. Correct Option: (b) Distributive Law of Multiplication over Addition

Proof: The algebraic expression $x \times (y + z) = (x \times y) + (x \times z)$ is the formal statement of the Distributive Law of Multiplication over Addition. Here, $x = -8$, $y = 5$, $z = -3$.

9. Correct Option: (b) $a - b$

Proof: Let us analyze each expression given that $a < 0$ and $b > 0$:

- (a) $b - a$: Since $a < 0$, $-a > 0$. Thus, $b - a$ is the sum of two positive numbers, which is always positive.
- (b) $a - b$: This can be written as $a + (-b)$. Since a is negative and $-b$ is negative, their sum must always be negative.
- (c) $a^2 + b$: Since $a \neq 0$, a^2 is always positive. The sum of positive numbers a^2 and b is positive.
- (d) $|a| - b$: This depends on their magnitudes. If $|a| > b$, the result is positive. If $|a| < b$, the result is negative.

Thus, $a - b$ is the only expression that is guaranteed to be negative in all cases.

10. Correct Option: (d) Both (a) and (c)

Proof: Since a and b are integers and their product $a \times b = -12$ is negative, one integer must be positive and the other negative. Let us list all integer factors of -12 and check the condition that $a + b > 0$:

- $a = 12, b = -1 \implies a + b = 11 > 0$ (Valid). Here, $a - b = 12 - (-1) = 13$.
- $a = 6, b = -2 \implies a + b = 4 > 0$ (Valid). Here, $a - b = 6 - (-2) = 8$.
- $a = 4, b = -3 \implies a + b = 1 > 0$ (Valid). Here, $a - b = 4 - (-3) = 7$.

(Reversing the assignments of a and b gives the same set of values with opposite signs, which violates $a + b > 0$). Thus, the possible values for $a - b$ are 7, 8, or 13. Since both 7 (option a) and 13 (option c) are valid, the correct choice is (d).

Solutions for Section B: Subjective Problems

1. **Solution:** The given expression is:

$$(-24) + 45 - (-16) - 13 + (-8)$$

First, eliminate double negatives and group positive/negative terms:

$$= -24 + 45 + 16 - 13 - 8$$

Group positive terms together and negative terms together:

$$\text{Positive terms: } 45 + 16 = 61$$

$$\text{Negative terms: } (-24) + (-13) + (-8) = -(24 + 13 + 8) = -45$$

Combine the results:

$$= 61 - 45 = 16$$

Final Answer: 16

2. **Solution:** The given expression is:

$$|-45| \div |-9| \times (-3) - |12|$$

Evaluate absolute values first:

$$|-45| = 45, \quad |-9| = 9, \quad |12| = 12$$

Substitute the values back:

$$= 45 \div 9 \times (-3) - 12$$

Apply DMAS order of operations:

- Division: $45 \div 9 = 5$. The expression becomes: $5 \times (-3) - 12$.

- Multiplication: $5 \times (-3) = -15$. The expression becomes: $-15 - 12$.
- Subtraction: $-15 - 12 = -27$.

Final Answer: -27

3. Solution: Given list of integers: $-18, 12, -4, 0, -25, 9, -1$

- (a) Grouping negative integers: $-25, -18, -4, -1$. Since $-25 < -18 < -4 < -1$, we order them. Zero separates them from positive integers: 0 . Grouping positive integers: $9, 12$. Since $9 < 12$. Combining all in ascending order:

$$-25, -18, -4, -1, 0, 9, 12$$

- (b) The smallest integer is -25 (first term) and the second-smallest is -18 (second term). All integers strictly between them are:

$$-24, -23, -22, -21, -20, -19$$

4. Solution: We need to find the product of the predecessor of -12 and the successor of -7 .

- **Predecessor of -12 :**

$$\text{Predecessor} = a - 1 = -12 - 1 = -13$$

- **Successor of -7 :**

$$\text{Successor} = a + 1 = -7 + 1 = -6$$

- **Product:**

$$\text{Product} = (-13) \times (-6)$$

Multiplying two negative numbers yields a positive product:

$$\text{Product} = +(13 \times 6) = 78$$

Final Answer: 78

5. Solution: We are given $a = -6, b = -4, c = 2$. We need to verify: $a \times (b + c) = (a \times b) + (a \times c)$.

- **Left Hand Side (L.H.S.):**

$$\text{L.H.S.} = a \times (b + c) = -6 \times (-4 + 2)$$

Calculate sum in brackets first: $-4 + 2 = -2$.

$$\text{L.H.S.} = -6 \times (-2) = 12$$

- **Right Hand Side (R.H.S.):**

$$\text{R.H.S.} = (a \times b) + (a \times c) = \{-6 \times (-4)\} + \{-6 \times 2\}$$

Calculate individual products first:

$$-6 \times (-4) = 24$$

$$-6 \times 2 = -12$$

Add the products:

$$\text{R.H.S.} = 24 + (-12) = 24 - 12 = 12$$

Since $\text{L.H.S.} = \text{R.H.S.} = 12$, the Distributive Law of Multiplication over Addition is verified.

6. Solution: We represent temperatures and changes using signed integers.

- Starting temperature at 6:00 AM: $T_0 = -5^\circ\text{C}$
- Rise in temperature (positive change): $+12^\circ\text{C}$
- Temperature at 2:00 PM: $T_1 = -5 + 12 = +7^\circ\text{C}$
- Fall in temperature (negative change): -14°C
- Temperature at midnight: $T_2 = 7 - 14 = -7^\circ\text{C}$

This can be written in a single equation:

$$T_{\text{final}} = -5 + 12 - 14 = 7 - 14 = -7^\circ\text{C}$$

Final Answer: The recorded temperature at midnight was -7°C .

7. Solution: We represent profit as a positive integer (+) and loss as a negative integer (-).

- Profit per pen: +Rs. 5
- Profit from selling 80 pens: $80 \times 5 = +\text{Rs. } 400$
- Loss per pencil: -Rs. 2
- Loss from selling 120 pencils: $120 \times (-2) = -\text{Rs. } 240$
- Net Transaction value:

$$\text{Net} = \text{Profit from pens} + \text{Loss from pencils}$$

$$\text{Net} = 400 + (-240) = 400 - 240 = +160$$

Since the final sum is positive, the vendor made a net profit of Rs. 160. **Final Answer:** Net Profit of Rs. 160.

8. Solution: Let the unknown integer be x . We are given:

$$\text{One integer} = -9$$

$$\text{Product} = -144$$

The algebraic equation is:

$$-9 \times x = -144$$

To find x , divide both sides by -9 :

$$x = -144 \div (-9)$$

Dividing two negative integers yields a positive quotient:

$$x = +(144 \div 9) = 16$$

Final Answer: The other integer is 16.

9. Solution: Given $|x| \leq 4$ and $|y| \leq 3$. This means:

$$x \in \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$$

$$y \in \{-3, -2, -1, 0, 1, 2, 3\}$$

(a) **Maximum possible value of $x - y$:**

To make $x - y$ as large as possible, we must select the largest possible value for x and the smallest (most negative) possible value for y .

$$x_{\text{max}} = 4, \quad y_{\text{min}} = -3$$

Substitute these into the expression:

$$(x - y)_{\text{max}} = 4 - (-3) = 4 + 3 = 7$$

(b) **Minimum possible value of $x \times y$:**

To minimize the product, we want the product to be as negative as possible. This occurs when we multiply the largest positive value from one set by the most negative value from the other set.

- Case 1: Select $x = 4$ (max positive) and $y = -3$ (min negative) $\implies 4 \times (-3) = -12$.
- Case 2: Select $x = -4$ (min negative) and $y = 3$ (max positive) $\implies -4 \times 3 = -12$.

In both scenarios, the absolute minimum product value is -12 .

Final Answer: (a) Maximum difference is 7, (b) Minimum product is -12 .

10. Solution: Let us establish our reference frame: sea level is represented as 0 meters. Heights above sea level are positive, and depths below sea level are negative.

(a) **Depth equation:**

The submarine starts at a depth of -45 meters. It ascends at a rate of $+5$ meters per minute. After t minutes, its position $d(t)$ in meters is given by:

$$d(t) = -45 + 5t$$

(b) **Time to reach -15 meters:**

Set $d(t) = -15$ and solve for t :

$$-15 = -45 + 5t \implies -15 + 45 = 5t \implies 30 = 5t$$

$$t = 30 \div 5 = 6 \text{ minutes}$$

It will take exactly 6 minutes for the submarine to ascend to a depth of -15 meters.

(c) **Absolute vertical distance:**

The explorer is on a platform at $+12$ meters. The submarine is at -15 meters.

The vertical distance between them is the difference of their positions:

$$\text{Distance} = \text{Platform position} - \text{Submarine position}$$

$$\text{Distance} = 12 - (-15) = 12 + 15 = 27 \text{ meters}$$

Final Answer: (a) $d(t) = -45 + 5t$, (b) 6 minutes, (c) 27 meters.